

AI Infrastructure Space – Astera Labs, SiTime & Credo Technology – Former Business Executive at Astera Labs Inc

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Specialist Background

- ▶ Over 25 years' experience in the data centre and semiconductor industry, covering product technical design and sales
- ▶ Extensive knowledge of the PCIe (peripheral component interconnect express) retimer landscape, AEC (active electrical cable) technology, MEMS's (micro-electro-mechanical systems) impact and the emerging shift toward CPO (co-packaged optics)
- ▶ Unique perspectives on client preferences, such as dual-sourcing trends at Amazon, growth trajectories for key components in AI clusters, plus the long-term strategic outlook for component vendors in a rapidly evolving AI hardware ecosystem

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Transcription begins

Analyst 1:

We are interested in companies such as SiTime, Astera Labs and Microchip. We may start with Astera Labs. We understand what it is doing, and it looks like it is benefiting a lot. What are your perspectives on its products and competitive landscape?

Specialist:

I can start with Astera. Astera is an execution machine. They have an amazing team. The founders are great. They all have their specialties and they all stay in their own wheelhouse. They have a lot of trust among the three of them. I don't know if you know their background, but they all came from TI, where they started the FTP business together or product line together, which is hugely successful for TI. They had a long history of working together, and that really shows in the way that they have built this company basically overnight, if you really think about it. A highly executive team there. They are constantly looking for ways to bring in good talent and continue this momentum that they have. As far as their products go, they really just know what they're doing. Again, the next level down of leadership they've brought in are just real experts. I think, on a competitive landscape point of view, with their product line, now they have this whole, like you buy all of their chips together, and they all work so seamlessly. It's like you just get a lot of benefit from the seamlessness among their chipsets and how they all work together. That was starting with the retimer chip. I have a couple of different thoughts here.

The retimer – what makes it so special is a lot of their firmware software, and they really focus on that. Casey is the mastermind behind that part of it. He's just a brilliant technologist. They really got their claws into a lot of the designs, being the first to market on PCIe Gen5 retimers. The way a lot of these hyperscalers design are cut-and-paste model. One piece is the hardware piece, which they've executed on. The other piece, the sticky part of it, is the software piece. They've really made their products very sticky through the software piece. That was a genius move right there, because, again, once you get integrated, there's a lot of cut-and-paste. You can cut-and-paste on the hardware side, all the traces and whatnot, but then also once you get that software into their software stack, it's almost impossible to get rid of them. Unless they do a massive overhaul of everything. If someone would want to take them out, which I don't know why they would, but in a case where they would want to take them out, it would take a very long time to get them out, because how sticky their technology is, both on the hardware side and the software side, and because these chips aren't MSA, multi-source agreement, like an optical module would be, you're in there for the long haul.

Analyst 1:

As you were saying, as Astera Labs loses share to Marvell, is that a big deal? It had almost 100% market share, growing more than 100%. Do you feel the growth is going to slow down dramatically because of that? What's your sense of the outlook for the next few years?

Specialist:

A couple of things. One, when you move to PCIe Gen6, there's just going to be a bigger TAM, because the data rates are doubling. Then there's just more need for retimers. The bucket is getting bigger, yes, they'll lose market share, but the TAM is getting bigger. I think that will probably be a wash. Where I think they're going to continue, and the revenue shows this as more of these other technologies come online, the AECs, or I think they're calling something different now. The cables, the active electrical cables and also the switching, they're just going to continue to gain market share. The switching market, in particular, that is dominated by Broadcom. Basically, Broadcom owns 100% of that market. The fact that they're able to come in and get share from Broadcom is huge. They're going to take share from Broadcom on the switching. Broadcom will take margin from them on the retimers. AEC is a crap shoot. Everybody and their brothers doing that. I just whatever with that stuff.

Analyst 1:

Is the AEC [active electrical cable] what Credo does?

Specialist:

Credo and then Molex do some of it, but it's a convoluted supply chain. As far as in the past, Credo was doing all vertical, and I heard some of their customers had bad experiences with them and so started to branch out. I think they started selling their chip only. I'm not sure, but I think they did, because their business model was to be vertically integrated, whereas Asteras was not. Asteras was to sell the little paddle cards and then go to a Molex or an Amphenol or whatever those guys and partner with those guys, because they know the copper cabling part, there's no margin in that, and it's just going to drop their chip/like module margin on the copper side.

Then I believe now they sell the chips, because the chips have the highest margin. Then a board is lower than that and or a module, whatever you want to call it. Then, the margin on a full cable assembly is [low]. I think, both Credo and Astera had to change their business model on the AECs to just sell chips, just to keep their margins higher. Credo is one, but then, Inphi also sells chipsets. You've got Inphi and probably Credo and Astera all selling chips into Molex and Amphenol, a TE, Volex, I don't know where they are right now. The cable vendors will have lower margins, but they can keep the high margins of the semiconductor chip.

Analyst 1:

Were you at the AWS event? Some news came out that Amazon was looking to either go away from Astera Labs or look at a dual source. Have you heard of that?

Specialist:

I haven't heard that. To be honest, I wasn't really looking into it. Would I be surprised if AWS wants to do a double source? No, I wouldn't. The place that Astera lacks on the switching side are the number of ports. In my knowledge, and granted, my knowledge is indifferent, it's still CSPs, but different CSPs. I'm aware of more of a need for higher port counts. They mentioned this, I think the higher port counts are for scaling up, whereas what they've been doing is scaling out. If they're able to hit those higher port counts, which they claim in their latest earnings calls, they're going to be going after the scaling up; that's where the big money is. That's directly taking away from Broadcom. Yes, directly taking away from Broadcom, because Broadcom's Tomahawk chip is just dominating the market.

Analyst 1:

Can you give the same ranking for Credo?

Specialist:

I don't know as much about Credo. Are we just talking about where they compete? Is it Credo vs these

guys?

Analyst 1:

If you know their market and the whole AEC market.

Specialist:

Personally, I think the AEC market is a dog. It's niche, it's limited, too many people have been burned in the past from them. I think the AEC market will go up, but it's not going to go up that much. I don't know. I just don't see the long life for the AECs. I've never been a big fan of the AEC market.

Analyst 1:

Could you do the same analysis on SiTime?

Specialist:

SiTime, now that they've entered the resonator market, that's a huge market. They had rumblings of that back when I was there 5-6 years ago. I knew that once they got into the resonator market, it would massively increase their revenue. I'm not clear on what the margins are on that, but I would imagine the margins are pretty good on just the resonator only, much higher than you would expect for other OXOs and those on the chip level. I think they're going to take off. Also, as far as data centre, as far as their AI stuff goes, I think they're going to take off, because really the MEMS solution is so much better, in my opinion, than Quartz for many reasons. I think they're going to continue to be on a tear as well. I don't really see them slowing down, especially with this new resonator market.

Analyst 1:

Give me a number, 0-10 on SiTime.

Specialist:

I'm going to go more like a 6.5-7. I'm a little less convinced of them over Astera. Just because there are competing solutions and they have drawbacks and stuff like that. As far as MEMS technology goes, they are going to kill it. Just Microchip in general, they've lost their way in the data centre, which is going to be massive growth. From a technology perspective, SiTime has the best technology when it comes to MEMS. However, Quartz is still a viable technology, still one that can keep going, and already is in the resonator market, obviously. I think there's more competition from different technologies, whereas on the Credo, Astera side, everybody has the same technology. It's just how they're implementing it.

Analyst 1:

We heard that there is no competition for SiTime, right?

Specialist:

That's not true.

Analyst 1:

Of course, for the older, the crystal technology, there is competition. However, I think on the MEMS side, it seems like there is no competition, is there?

Specialist:

Yes, correct. On the MEMS side, there is no competition.

Analyst 1:

Shouldn't everything go to MEMS [micro-electro-mechanical system]? Let's say, in five years, don't

you feel that MEMS, as a better technology, should hopefully just keep growing?

Specialist:

Yes. There are some weaknesses, some pretty profound weaknesses like temperature. If MEMS can correct or maybe correct isn't right, get around some of the inherent flaws with MEMS, then yes, SiTime will explode. It's just whether or not they can get around those. There are some pretty key issues with MEMS at temperature; they have these blackout spots where the clock just goes cuckoo. I feel like there's probably a little bit more life left in crystals vs MEMS. That's the question is, is MEMS going to get so much better? If MEMS get so much better, then yes, they will dominate because there is no competition, zero competition for SiTime. Do you believe MEMS is going to be able to solve these inherent issues that they have? I would have to look into more if they've been able to solve those issues.

Analyst 1:

Do you feel that SiTime's growth is going to slow down because it has picked the low-hanging fruits, and there will be no more growth? Do you feel the growth is going to continue because more and more people are going to use MEMS?

Specialist:

Their growth is going to continue because they've gotten into a new market, the resonator market.

Analyst 1:

Is the growth going to continue because of the MEMS resonator market?

Specialist:

The resonator market, it just opened up a whole new SAM/TAM for them that they weren't a massive new industry, I don't want to say industry, but a massive new SAM/TAM that they were never able to play in before. That's where most of their growth is coming from. Also, MEMS is better suited for AI technology for sure, like the data centres. It's like those two things. The resonator market is huge.

Analyst 1:

That's pretty big, yes.

Do you feel that there's a lot of need for SiTime in the AI field, including AI chips and AI data centres? How do you think about that market?

Specialist:

There is a need for SiTime as data rates increase. I would say, yes, they are needed in the data centre, but they are also needed in the automotive industry. That's where those guys will really shine for sure because our cars are computers now. They're mostly data centres. They're taking all the information from the sensors and blah, blah, blah and processing all that, especially with autonomous vehicles, that's where they're going to see a lot of growth because the MEMS is inherently resistant to physical shocks. Anything that's data centre or data centre-like will require MEMS, yes. I wouldn't strictly limit it to data centres because there are other things. I mentioned robotics. Robotics is going to need MEMS-based timing, too.

Analyst 1:

Yes, I feel that's pretty far in the future.

Specialist:

I'm just saying it's not strictly held to data centres. It's anything that requires high-accuracy timing at

high speeds that also have mechanical shock to it. That's where they're going to really shine. I guess I'm saying don't limit yourselves just to the data centre.

Analyst 1:

We've been hearing about CPO [co-packaged optics], but it seems that people are saying CPO won't happen for at least three years. What do you think of the next two years?

Specialist:

That's a hard one because they're both totally separate. If SiTime can really get into this resonator market and unlock that, that's going to be a windfall for them because it's a massive market that they've never touched before. If they can figure out the recipe and make all these custom resonators, that's going to be huge for them. Astera, I'll be interested to see how much the switch market is another huge market. If they're able to penetrate the switch market, that's going to be a huge win for them. I don't know what Credo is doing anything. They're not doing anything different or innovative. I think they're getting into more ethernet, but ethernet is so saturated. Sorry, I think they're getting into retimers or looking into retimers and that market is so saturated. I would bet on either Astera or SiTime over Credo for sure. Then, between the two of them, it's just hard. They're just like in such different industries.

Analyst 1:

Is it just because all the cables are getting into CPO and optics? Which vendors are going to benefit from that optics growth? We know some of the transceiver companies, such as Coherent and Lumentum. Are there any other companies, such as players in the components or subcomponents field, or any field that you find interesting?

Specialist:

I'm hearing there is a laser shortage right now. I don't know what specific lasers. I think people that control the lasers are going to have benefit. At the higher data rates, people that do silicon photonics chips, but those aren't going to be public. Those are all going to be private. Those are all the private guys. Coherent and Lumentum are probably the biggest players in that whole field. I can't, off the top of my head, think of someone else that I would. I think II-VI has fallen behind. I haven't heard anything from them.

Analyst 1:

Are there other component players? Microchip seems mostly an analogue player, so maybe it doesn't really benefit directly, right?

Specialist:

Microchip? For the AI bubble? On the infrastructure side, no. They're not. I would not put any money into Microchip. Not for AI. I don't know enough about their... I was there for their AI stuff and for their data centre stuff. I did mostly data centre stuff for them. No, they're not going to be competitive.

Analyst 2:

You brought up some of the ASIC [application-specific integrated circuit] programmes. Is SiTime in any of those programmes, such as the Google TPU [tensor processing unit] and the AWS Trainium? Anything that you know?

Specialist:

On a resonator side?

Analyst 2:

Or MEMS side, yes.

Specialist:

On a TPU board, you mean?

Analyst 2:

Yes.

Specialist:

I don't know. I wouldn't be surprised, though. It could unlock a whole bunch of resonator stuff. I don't know if those chips have the timing on the inside or the outside.

Analyst 2:

Does that make a difference?

Specialist:

Yes. Resonators go inside a chip, and then timing devices go on the outside of a chip on a board. Obviously, the resonator would go inside a TPU chip, whereas an oscillator would go on the outside on a TPU board.

Analyst 2:

Is SiTime's data centre business mostly resonators today, or is it completely new?

Specialist:

That's completely new. They just announced it, so that's completely new. That's why I keep saying, if they can figure out the recipe for that, that's going to unlock an absurd amount of industry.

Analyst 2:

You had mentioned something along the lines of MEMS, there's a certain power threshold, and they sort of bug out. Our understanding was that, obviously, the new-generation GPUs or TPUs is consuming more and more power, and MEMS was better positioned to handle that increased power temperature, but it sounds like that's not the case.

Specialist:

They do great. There's less variation over temperature, but there are regions where they just die, not die, but they stop working at certain temperature profiles. If you look at the graph of the resonator, there are these – I can't remember what they call, I can't remember the technical name now – basically dead zones where they just stop working at a specific temperature range or something like that. Again, if they can get around them, they're not catastrophic issues. If they can get around some of these inherent obstacles with MEMS, then MEMS is a clear winner. I'd have to look more because I haven't looked into their technology.

Analyst 1:

I guess that whatever those hurdles have been for MEMS, they've been in the past. The company is growing pretty fast. In the future, would you think that there are more opportunities for MEMS or do you think there are less opportunities for MEMS?

Specialist:

No, more, more, more. 100%.

Analyst 1:

There would be some benefit of doing this in a MEMS way vs the traditional way, right?

Specialist:

100%. MEMS is a superior technology, a million percent.

Analyst 1:

Why do you say that the resonator is tricky?

Specialist:

Typically, a resonator is needed for a very specific frequency, and the recipe to get the right frequency can be a little bit of black magic. It's a science, but it's also a bit of art and magic. Now, if they're using AI to create these recipes and they've figured out how to create better recipes for the resonator and they can do it quickly, then that's great because each of these companies that are building like a TPU or things where they're building chips in the millions. If they can get the recipe done quickly and right for the resonator that goes inside those chips, then they will take that resonator business will take off. If it's essentially you're starting from scratch, each recipe you're starting from scratch for each different chipset, and it takes 1.5 years or whatever to develop each resonator for that specific chip or use case, then they're not going to be able to penetrate that market as fast. Does that make sense?

Analyst 1:

In the industry, I know Steve at Microchip used to be pretty strong. Who else is a rock star? Hock Tan?

Specialist:

That's a good question. I think Hock is a league of his own, to be honest. No one's as brutal as him that I'm aware of. There's probably some equivalent on some of the other technologies, but if you're talking strictly data centre, I'm don't know enough about like wifi and all those guys, I don't know. I imagine there's probably someone on that. These other technologies that I'm not as familiar with. I don't see anyone who's as brutal and as brilliant as Hock when it comes to business. A lot of people don't want to deal with Broadcom. I very much see any kind of ASIC being developed that more people would want to go to Marvell. For ASIC, it's only Marvell and Broadcom unless you're doing it internally. This has been rumoured for a long time. Everybody has been trying to displace Broadcom. It's been over a decade, but they just can't because their technology is just too good, but it's like Google.

Transcription ends

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